Finite permutation cores for the nonunital product question

A rigorous partial result for arXiv:1705.08967

Autonomous solve-first investigation, lane 12

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Abstract

Niemiec and Wójcik ask whether a finite product of weakly (respectively strongly) right amenable semigroups with topology (SGTs) remains right amenable when the factors need not be unital. We do not settle the general question. We give an exact finite-SGT reduction to a common permutation core, use it to exclude every counterexample whose factors have order at most three, and perform a complete self-product audit on 1,870 canonical order-four extension types. We also isolate why the usual iterated-mean proof breaks for arbitrary SGTs.

The source question. An SGT is a semigroup equipped with an arbitrary topology; multiplication is not assumed continuous. For $f_s(x) = f(xs)$, the paper defines $C_{\text{norm}}(S)$ and $C_{\text{weak}}(S)$ by requiring $f \in C_r(S)$ and continuity of $s \mapsto f_s$ and $s \mapsto f_{st}$ in the norm or weak topology, for every $t \in S$. Immediately after Proposition 5.1(E), the authors ask whether its unitality hypothesis can be dropped, even for a finite family [1, p. 14].

1 Exact finite reduction

Let S be finite. Write $x \sim y$ when x, y are in the same connected component of the finite topology. Components are clopen, so the continuous real functions are exactly those constant on the $\sim$-classes. Since every function space below is finite dimensional, weak and norm continuity coincide.

Define R to be the equivalence relation generated by the following pairs:

$$
\begin{aligned}
&x \; R \; y && (x \sim y),\\
&xs \; R \; ys && (x \sim y,\ s \in S),\\
&xs \; R \; xs' && (s \sim s',\ x \in S),\\
&x(st) \; R \; x(s't) && (s \sim s',\ x, t \in S).
\end{aligned}
$$

Proposition 1 (finite function quotient)**.** *$C_{\text{norm}}(S) = C_{\text{weak}}(S)$ is precisely the space of real functions constant on the R-classes. Moreover, R is stable under every right multiplication, so*

$$\rho_s : S/R \longrightarrow S/R, \qquad [x] \longmapsto [xs]$$

is well defined.

Proof. The first line is ordinary continuity. The second is exactly the condition $f_s \in C_b(S)$ for every s. A map from a finite space to a Hausdorff vector space is continuous exactly when it is constant on connected components; this turns the two translation-orbit conditions into the third and fourth lines. Thus the displayed relations are necessary and sufficient. Stability under right multiplication follows from associativity: the right translate of a relation in lines one through four is respectively a relation in lines two, two, four, and four. □

For maps ρ_s on a finite set Q, call $K \subseteq Q$ a *common permutation core* if $K \neq \varnothing$ and every ρ_s restricts to a permutation of K.

Theorem 2 (permutation-core criterion)**.** *A finite SGT S is weakly right amenable (equivalently, strongly right amenable) if and only if the right action $\{\rho_s : s \in S\}$ on S/R has a common permutation core.*

Proof. A mean on the finite-dimensional space of Proposition 1 is a probability vector p on $Q = S/R$. Right invariance says $(\rho_s)_* p = p$ for every s. For one deterministic map on a finite set, an invariant probability gives zero mass to every transient point. Hence each ρ_s permutes $\operatorname{supp} p$, proving necessity. Conversely, the uniform probability on a common permutation core is invariant under every ρ_s and therefore defines a right invariant mean. □

The largest possible core has a simple exact algorithm. Start with $K_0 = Q$. Given K_n, retain only points that lie on a cycle of every ρ_s contained in K_n, and call the result K_{n+1}. The process stabilizes in at most $|Q|$ strict deletions. Any common core survives every deletion, while a nonempty fixed point is itself a common core. This proves both correctness and termination without numerical tolerance.

2 Exhaustive and extended searches

Every set partition occurs as the component partition of a finite topology (take a disjoint union of indiscrete blocks), so enumerating partitions loses no finite cases relevant to these function spaces.

order	labelled semigroups	component partitions	amenable pairs
1	1	1	1
2	8	2	15
3	113	5	534

The enumeration tested every one of the 72,010 unordered pairs of nonunital amenable candidates in these rows. Every product had a nonempty permutation core. The core test was independently compared with a linear-program feasibility test on all 582 factor cases and 500 sampled products; all 1,082 answers agreed.

To push past order three, a local associative extension enumerator began with all 113 labelled order-three semigroups and exhausted the seven multiplication entries involving a fourth element. It produced 2,235 distinct labelled order-four tables containing a three-element subsemigroup. After testing all 15 component partitions and canonicalizing under all 24 relabellings, 1,870 amenable nonunital finite-SGT types remained. *Every one of their 1,870 self-products is amenable.* In addition, 10,000 random cross-pairs from the uncanonicalized catalog and 8,000 pairs from independently generated finite transformation semigroups produced no counterexample.

These computations are exhaustive only where explicitly stated. In particular, the cross-pair order-four search and the larger transformation search are evidence, not proofs.

3 Why the general proof still fails

For abstract amenable semigroups one writes

$$M(f) = m_S\bigl(s \mapsto m_T(t \mapsto f(s,t))\bigr).$$

Without units this formula need not be defined on the paper's function spaces. A slice $t \mapsto f(s,t)$ need not belong to $C_{\mathrm{norm}}(T)$ or $C_{\mathrm{weak}}(T)$: continuity of $t \mapsto f(s,tv)$ cannot be obtained from a product translate unless the first coordinate has an element that leaves s fixed. Extra right-factor anchors repair an individual slice, but after averaging there is no reason for the outer translation-orbit map to remain norm or weak continuous on a noncompact factor.

The finite criterion shows why the most obvious counterexamples also fail. Product functions can distinguish extra "indecomposable" layers, but every invariant probability assigns zero mass to layers transient under a right map. To prove the finite product theorem one would still need to lift the product of the two factor cores through the generally finer product function quotient. The searches support such a lift, but a general proof was not obtained.

Scope. The unrestricted nonunital product question remains open in this packet. The rigorous contribution is Theorems 1–2, the complete order-three exclusion, and a reproducible order-four self-product audit. Any future counterexample must evade all of these finite mechanisms; the most plausible remaining setting is an infinite noncompact SGT where averaging destroys translation-orbit continuity.

References

[1] P. Niemiec and P. Wójcik, *Applications of amenable semigroups in operator theory*, Studia Math. 252 (2020), 27–48; arXiv:1705.08967.